

8. Cystic fibrosis is an example of an inherited disease caused by a recessive allele. A carrier of the disease has a genotype Ff.

- (a) Compare the chances of a child being born with cystic fibrosis if both parents are carriers, with a child whose father suffers from cystic fibrosis and a mother who is a carrier. [4]

- (b) Cystic fibrosis causes a build-up of thick mucus in the lungs, which leads to poor gas exchange. Mucus in the lungs also allows increased bacterial growth that is responsible for infections.

Explain how the body defends itself against bacterial infection in the lungs. [3]

- (c) Sufferers of cystic fibrosis benefit from being physically active to slow the rate of decline in lung function.

Explain the short-term and long-term effects of exercise on lung function. [3]

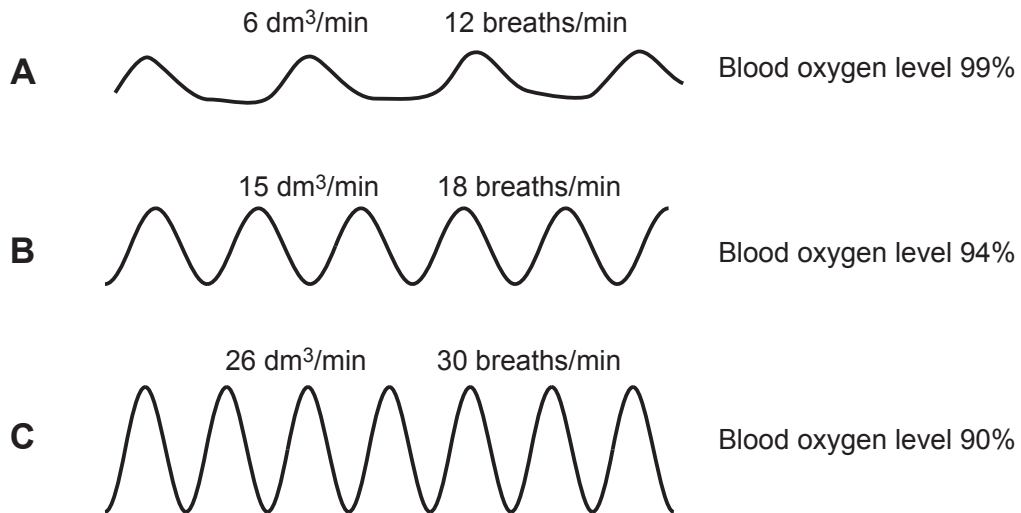


- (d) The diagram below gives information about the breathing of three different individuals, all of whom are resting.

A is a person who is not affected by cystic fibrosis.

B is a person with cystic fibrosis.

C is a person with cystic fibrosis and a lung infection.



The numbers show the volume of air breathed in per minute in dm³ per min and also the number of breaths per minute.

- (i) Use the equation:

$$\text{volume of air per breath} = \frac{\text{volume of air breathed in per minute}}{\text{number of breaths per minute}}$$

to compare the volume of air breathed in per breath for each of the breathing patterns.

[4]

A B C

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- (ii) Compare the number of breaths per minute with blood oxygen levels in the three individuals. [2]

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END OF PAPER

Examiner
only

16



Group

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1 H Hydrogen 1										4 He Helium 2									
7 Li Lithium 3		9 Be Beryllium 4		23 Na Sodium 11		24 Mg Magnesium 12		39 K Potassium 19		40 Ca Calcium 20		88 Sr Strontium 38		133 Cs Caesium 55		223 Fr Francium 87			
11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17			
27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18		70 Ga Gallium 31		73 Ge Germanium 32		84 Kr Krypton 36			
65 Zn Zinc 30		63.5 Cu Copper 29		59 Ni Nickel 28		59 Co Cobalt 27		56 Fe Iron 26		55 Mn Manganese 25		52 Cr Chromium 24		51 V Vanadium 23		48 Ti Titanium 22			
112 Cd Cadmium 48		108 Ag Silver 47		106 Pd Palladium 46		103 Rh Rhodium 45		101 Ru Ruthenium 44		99 Tc Technetium 43		96 Mo Molybdenum 42		93 Nb Niobium 41		91 Zr Zirconium 40			
115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72			
204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86		226 Ra Radium 88		227 Ac Actinium 89					
Key																			

relative atomic mass

A_r	Symbol	Name	Z
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